

CF00: Lifted-State Induced Geometry and Emergent Dynamics

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Abstract

CF00 derives induced geometry as the smooth quotient shadow of the finite lifted-state grammar already present in the VDM canon. The starting point is not the quantum-geometric tensor. The starting point is the CF000 invariant stack---non-discharge, same-axis saturation, orthogonal re-articulation, proto-number, algebra unlock, and the two-axis ORS algebra---together with the Phase Calculus carried state $\hat{X}_i=(A,q,\theta,\kappa,c)$, operator core Q,B,L , and exact quotient criterion $\Pi \circ E = G \circ \Pi$. From these primitives the paper derives the carrier domain calM , effective complex representative fibers, local phase redundancy, and horizontal quotient variation. The familiar quantum-geometric tensor then appears downstream as the unique minimal Hermitian covariance of horizontal lifted-state variation in the smooth representative shadow, with g and Ω as its symmetric and antisymmetric sectors. This reframing turns the exact identity $(g)/(\Omega)=1/4$ in the local chart into a checkable shadow of retained-state quotient geometry rather than an assumed starting formula. The dynamical half of the paper is correspondingly sharpened: induced geometry narrows admissible transport to vertical-blind horizontal motion and supplies the first scalar-generated $J+M$ architecture used by CF01 and later retained-state modules.

Void Dynamics Model , lifted state , quotient descent , induced geometry , quantum geometric tensor , emergent dynamics

1. Introduction and Main Thread

CF00 is the first smooth retained-state geometry paper in the VDM branch. Its job is to show how a familiar geometric object, the quantum-geometric tensor, becomes visible only after a deeper retained-state grammar has already done the load-bearing work.

The dependency order is explicit. CF000 supplies the primitive invariant stack: non-discharge, same-axis saturation, orthogonal re-articulation, proto-bbN, the 2D numerical relation grid, algebra unlock, and the imaginary unit as the algebraic witness of ORS. Phase Calculus supplies the exact carried object

$\hat{X}_i=(A,q,\theta,\kappa,c)$, $q=(u,v)$, $c=[\theta-\frac{1}{2}\pi, \theta+\frac{1}{2}\pi]$,

together with the finite operator core Q,B,L and the quotient law

$\Pi \circ E = G \circ \Pi$.

CF00 uses those ingredients directly. It does not begin with a Hilbert-space bilinear and then interpret the result back into VDM. It begins with retained lifted transport, derives the smooth carrier shadow, quotients vertical representative motion, and only then obtains the horizontal Hermitian covariance recognized as the QGT.

The guiding picture is simple. The lifted state carries what the visible representative forgets. The quotient projector decides which branch-visible shadow is being read. The smooth carrier branch is one such shadow. In that branch, local lifted-state variation becomes horizontal representative variation after pure phase motion is removed. The minimal object that keeps both magnitude and oriented ORS order of that horizontal variation is a Hermitian covariance tensor. Its real part is the induced metric g ; its imaginary part is the induced curvature/symplectic sector Ω .

This turns the paper's exact symbolic example into more than a classical re-derivation. In the chart

$|\psi(u,v)\rangle = \frac{1}{\sqrt{2}}(1 + u^2 + v^2)^{-1/2}(1 + iu + v) \rightarrow p$,

the quotient-induced tensor gives $(g)/(\Omega)=1/4$ exactly. The companion SymPy script reduces the projector, tensor, s

The dynamics half is also stage-local. CF00 does not claim to finish all runtime dynamics. It derives the first admissible transport architecture visible at this stage: vertical-blind horizontal motion, with reversible transport through the antisymmetric sector and irreversible transport through the symmetric sector. CF01 continues this into the full QGT-to-metriplectic bracket construction, while later Phase Calculus and quotient-descent modules retain the finite-state grammar underneath the smooth shadow.

@@HEADING1:2. Root Status, Inherited Stack, and Notation Convention@@

2.1 Stage status

CF00 is the first retained-state carrier-geometry Complete Formalism after CF000. CF000 closes the pre-carrier route through 2D and hands forward not a primitive carrier, but the post-2D invariant stack: the primitive bifurcation invariant, the 1D axis-proper effective invariant (continuous same-axis bifurcation, proto-bbN), and the 2D axis-proper effective invariant (the numerical relation-grid invariant together with algebra unlock). CF00 therefore functions as the first stage at which that inherited stack is used to derive the carrier domain and then the induced geometry and emergent dynamics built on that derived carrier.

Any downstream document in this branch that uses quotient-physical variation, induced geometry, scalar-generated mixed dynamics, constitutive reservoir closure, or support-derived locality inherits those structures from CF00.

2.2 Notation convention

Two different kinds of object must not share the same symbol.

-
calM denotes the carrier domain derived at the CF00 stage:
the arena on which the representative state-family is defined.

-
M denotes the dissipative symmetric operator or sector in the
emergent dynamical split $J + M$.

This distinction is ontological, not cosmetic.

-
calM is a carrier arena.

-
M is an emergent operator on quotient cotangent data.

No statement in this document uses plain M for the carrier domain.
All carrier-domain quantities use calM and derived notation
such as d_{calM} for the carrier-domain distance.

2.3 The stage question

The stage question for CF00 is:

Given the CF000 invariant stack and the Phase Calculus lifted-state grammar, what smooth carrier shadow is forced, and what local geometry appears when retained lifted transport is read through that shadow after vertical representative motion has been quotiented out?

The answer of CF00 is:

The inherited stack and lifted-state grammar force a retained-first descent chain. The carrier calM , complex representative fibers, local $U(1)$ redundancy, and horizontal quotient projection are derived before any geometry is named. The quantum-geometric tensor then appears as the smooth-shadow Hermitian covariance of horizontal lifted-state variation. Its sectors (g, Ω) are therefore downstream quotient geometry, not primitive input.

2.4A A0-closure source map

A0-closure in CF00 means that no object may function as a hidden primitive of the VDM branch. The revised source map is therefore stricter than a provenance table: it constrains what is allowed to carry theorem burden.

p0.27p0.66

-> prule

Source class | Role in CF00

CF000 primitive stack | Non-discharge, same-axis saturation, orthogonal re-articulation, proto-number, algebra unlock, the two-axis ORS algebra, and the rule that unresolved burden must remain borne under continued articulation.

Phase Calculus lifted grammar | The carried state $\text{hatXi}=(A,q,\theta,\kappa,c)$, the finite operator core Q,B,L , selector-closed evolution where needed, exact return criteria, and the quotient law $\Pi \circ E=G \circ \Pi$.

CF00 derived smooth shadow | The carrier domain calM , smooth representative projection, normalized complex fibers, local phase redundancy, vertical/horizontal splitting, and the horizontal covariance tensor.

Downstream quotient geometry | The QGT formula, g , Ω , the determinant identity in Section 5.6, vertical-blind transport, and the scalar-generated $J+M$ architecture. These are outputs of the lifted-state-to-smooth-shadow descent, not starting primitives.

Forward integration | CF01 supplies the full QGT-to-metriplectic bracket formalism; CF03, the Infinity Resolution branch, Farey remainder recursion, Transcendental Wall, Operational Utility, and later Phase Calculus modules provide hierarchy, retained-germ, multiscale, and branch-translation continuations of the same retained-first method.

2.4B Lifted-state descent spine

Theorem 2.4B.1 (Retained-first geometry descent).

The smooth induced geometry used in CF00 is a quotient shadow of the lifted-state grammar. Its dependency chain is

CF000 invariant ->

hatXi ->

Q,B,L ->

$(\Pi \circ E=G \circ \Pi)$ ->

smooth horizontal covariance ->

(Q,g,Ω) .

Consequently, the QGT is not the first object of the construction. It is the first smooth Hermitian covariance visible after exact lifted transport has been projected to a normalized complex representative branch and vertical phase motion has been removed.

Status. Proved in this CF from inherited canon interfaces.

Proof. CF000 provides the invariant that must remain borne. Phase Calculus provides the smallest exact carried state i

@@@HEADING2:2.4 Inherited stack and stage

scope@@

Theorem 2.4.1 (Inherited invariant stack and derivation scope).

Within CF00, the post-2D invariant stack delivered by CF000 is the inherited primitive input of this stage. That stack consists of: the primitive bifurcation invariant (unresolved two-pole opposition), the 1D axis-proper effective invariant (continuous same-axis bifurcation, proto-bbN), and the 2D axis-proper effective invariant

(numerical relation-grid invariant, algebra unlock). CF00 does not reopen the pre-carrier origin burden. Its formal burden is to derive, from that stack, the first lawful host/carrier structure and then the quotient-physical geometric and dynamical structures forced by representative variation on that derived carrier.

Status. Proved in this CF.

Proof. CF000 closes the
 0D -> 1D -> 2D route and delivers:
 proto-bbN at 1D, ordered-pair algebra and bbC (as
 ORS expressed algebraically) at 2D, and the explicit handoff requirement
 that CF00 must derive calM rather than assume it. CF00
 therefore begins from the inherited stack as given input and derives, in
 strict order:

enumi.

- bbR from proto-bbN via the standard algebraic
completion chain;
- bbC from the 2D ORS rotation operator already licensed by
CF000 Section 7.4.1;
- topology on the completed number structure;
- the continuous completion of the discrete ORS quarter-turn on the
two-axis carrier implied by the real completion and the topological
structure;
- the primitive half-turn constant π as the unique least positive
opposite-pole attainment of that continuous ORS completion;
- differentiability as lawful variation on the topological structure;
- the carrier domain calM as the first admissible host of
continued invariant-bearing articulation past 2D;
- the effective Hilbert bundle over calM using bbC
from step 2;
- local U(1) redundancy from the complex phase structure acting on
normalized fibers.

Only after step 7 does the representative ontology of Section 3 become valid.
 The formal burden of the pre-carrier origin belongs to CF000.

@@HEADING2:2.5 Derivation of the carrier domain from the inherited
 invariant
 stack@@

2.5.1 From proto-bbN to
 bbR2.5.1 From proto- mathbbN to mathbbR

Theorem 2.5.1 (bbR as forced completion of proto-bbN).

The proto-natural-number structure licensed at 1D by CF000 generates, under successive forced completions, the real number field bbR.

Status. Proved in this CF.

Proof. Four completion steps are forced by the same saturation logic that forced 1D $\rightarrow$ 2D in CF000.

-

bbN $\rightarrow$ bbZ: Lawful binary combination, licensed at 2D, must be closed under the inverse of addition. If $a-b$ is not always admitted, the set is not saturated with respect to lawful combination. Therefore the integers bbZ are forced as the closure of proto-bbN under additive inversion.

-

bbZ $\rightarrow$ bbQ: Multiplication is licensed as the second lawful binary combination on the two-axis ordered-pair structure (Theorem 4.9.1 of CF000). Its inverse, division, is then forced by the same closure logic. Closure under multiplicative inversion, where the divisor is nonzero, forces bbQ.

-

bbQ $\rightarrow$ bbR: The rationals contain Cauchy sequences that do not converge within bbQ. A system that admits lawful sequential structure but contains non-terminating sequences without limits is not same-axis saturated: those limit points are lawful articulations that remain unborne. Cauchy completion therefore forces bbR.

Thus the real number field bbR is the first complete ordered field derivable from the CF000 handoff. No external postulate is required.

2.5.2 bbC as the 2D ORS

algebra 2.5.2 $\mathbb{bbC}$ as the 2D ORS algebra

Theorem 2.5.2 (bbC as licensed by CF000).
The complex number field $\mathbb{bbC} = \mathbb{bbR} + i\mathbb{bbR}$ is the natural algebra of the two-axis invariant-bearing structure admitted at 2D. The imaginary unit i is the algebraic form of the orthogonal rotation operator ORS, satisfying $i^2 = -1$ because returning along the orthogonal rotation path reaches the opposite pole of the invariant.

Status. Proved in this CF.

Proof. CF000 Section 7.4.1 proves this directly. The rotation operator R satisfying $R(1,0) = (0,1)$ and $R(0,1) = (-1,0)$ has $R^2 = -1$ because the two poles are irreconcilable. This is exactly i . The field $\mathbb{bbC} = \mathbb{bbR}[i]/(i^2+1)$ is therefore inherited from the CF000 closure, not postulated.

@@HEADING2:2.5.3 Topology from ordered completion@@

Definition/Theorem 2.5.3 (Topology forced by ordered completion).
The ordering structure of bbR together with the completion metric $d(a,b) = |a-b|$ generates a natural topology on bbR via the open-ball neighborhood structure. The product topology on $\mathbb{bbR}^n$ for finite n follows. This is not an external import: it is the lawful closure/completion burden expressed as topological structure.

Status. Derived structural consequence.

Proof. Once the real completion is present, open neighborhoods are fixed by the induced metric and ordered-field compatibility. No additional primitive topological axiom is required to specify which local neighborhoods count as admissible. The topology is the one already determined by ordered completion and the completion metric.

2.5.3b π as the primitive half-turn constant of the ORS orbit

Theorem 2.5.3b.1 (Continuous completion of ORS).

Once the 2D ORS quarter-turn of CF000, the real completion of Theorem 2.5.1, and the topological continuity licensed b

$$R_0 = I,$$

$$R_{\theta} = R_{\theta} R_0,$$

and extending the already-earned discrete quarter-turn action.

Status. Derived structural consequence.

Proof. CF000 closes the discrete orthogonal re-articulation step and yields the quarter-turn operator whose square is

$$R^2 = -I.$$

Theorem 2.5.1 supplies the completed ordered field needed for continuous parameterization, and Theorem 2.5.3 supplies

Definition 2.5.3b.2 (Primitive half-turn constant).

Let R_{θ} be the continuous ORS completion on the two-axis carrier. Define

$$\pi := \theta > 0 : R_{\theta} = -I.$$

This is the first positive rotation parameter at which the invariant reaches its opposite pole. It is the minimal angul

Theorem 2.5.3b.3 (π as the unique least positive half-turn).

The quantity π defined above is the unique least positive parameter at which the ORS orbit reaches the opposite pole.

Hence π is not primitively imported from circle-area, trigonometric series, or external convention. It is first earned as the half-turn constant of the invariant's own continuous rotational expression.

Status. Derived structural consequence.

Proof. Because the discrete ORS seed satisfies $R^2 = -I$, two successive quarter-turn applications reach the opposite pole

$$H := \{\theta > 0 : R_{\theta} = -I\}$$

is nonempty. Also $R_0 = I \neq -I$. By continuity of R_{θ} at 0, there exists $\epsilon > 0$ such that $R_{\theta} \neq -I$ for every

$$H \subset [0, \infty),$$

so the opposite-pole hit set is bounded away from 0. By continuity of the completed orbit, H is closed in $(0, \infty)$. S

Corollary 2.5.3b.4 (Geometric realization of the half-turn constant).

Under the continuous ORS completion, the orbit of a unit carrier point is the unit circle, the region it encloses is t

$$\pi = \int_0^1 dx dy,$$

$$D^2 = (x, y) : x^2 + y^2 \leq 1.$$

Status. Derived geometric corollary.

Proof. Once the ORS action is continuously completed on the two-axis carrier, the orbit of a unit carrier point closes into the unit circle. The enclosed region is the unit disk. The geometric measure constant attached to that disk is not a new primitive definition; it is the geometric realization of the already-earned half-turn constant.

Remark. This subsection does not yet claim transcendence. Its completed claim is narrower and structurally earlier: CF

@@@2.5.4 Differentiability from lawful

variation@@@

Definition/Theorem 2.5.4 (C^1 differentiability as forced variation structure).

On a topological carrier equipped with the real number field, a local variation structure is a map that assigns to each point a linear approximation of the change in a field value under small displacement. Requiring that this approximation be well-defined and consistent --- that the error in the linear approximation goes to zero faster than the displacement --- is exactly C^1 differentiability.

Status. Derived structural consequence.

Proof. The invariant cannot discharge, so local variation cannot be arbitrary if continued articulation is to remain lawfully comparable from point to point. The weakest regularity that keeps local variation well-defined, linearly comparable, and stable under the quotient-physical constructions that follow is C^1 . Anything weaker does not support a consistent projected-variation calculus; anything stronger is not yet forced at the minimum stage of CF00. For the downstream QGT-to-metriplectic bracket program of CF01, however, the projected variation constructions of Section 4--5 must be iterated to higher order. The same forcing argument that requires C^1 --- invariant-bearing local variation must be linearly comparable --- applies recursively: second-order variation must be linearly comparable to first-order variation, and so on. This forces C^∞ smoothness as the saturation of the differentiability burden in the presence of iterated variation. CF00 operates at C^1 as the minimum; CF01 may stipulate C^∞ as the iterated completion of that minimum, which is licensed by the same invariant logic applied recursively.

2.5.5 The carrier domain calM as derived

host 2.5.5 The carrier domain mathcalM as derived host

Theorem 2.5.5 (calM as the first admissible host of continued invariant-bearing articulation past 2D).

Given bbR (Theorem 2.5.1), bbC (Theorem 2.5.2), topology (Theorem 2.5.3), the primitive half-turn constant π (Theorem 2.5.3b.3), and C^1 differentiability (Theorem 2.5.4), the carrier domain calM is defined as a connected C^1 domain admitting local coordinates in bbR^n for some finite n .

calM is not assumed. It is the first structure that can host continued invariant-bearing articulation past the 2D algebraic stage: it has a real number field for coordinates, a complex field for the representative state fiber, a topology for local variation to be well-defined, the primitive half-turn constant π already fixed by the continuous ORS orbit, and differentiability for the projected variation construction of Section 4 to proceed.

Status. Proved in this CF.

Proof. Nothing weaker than a connected C^1 coordinate domain can host the local representative variation, local continuity, and projected derivative constructions used later in the paper. Nothing richer is yet forced at stage entry. The first admissible host is therefore the connected C^1 carrier domain calM built on

the already earned number, topological, and variation structure. It is input for derivations in Section 3 onward only in the sense that it is now earned before being used.

2.5.6 Hilbert bundle from calM

and

bbC 2.5.6 Hilbert bundle from mathcalM and mathbbC

Theorem 2.5.6 (Effective Hilbert bundle as derived structure).

Given calM (Theorem 2.5.5) and bbC (Theorem 2.5.2), the effective complex Hilbert bundle $\pi: \text{calH} \rightarrow \text{calM}$ is the minimal fiber bundle over calM with fibers carrying inner-product structure over bbC sufficient to define normalization, overlap, projected variation, and local phase redundancy.

Status. Proved in this CF.

Proof. The quotient-physical constructions of Section 4 require a normalized representative state at each point of the carrier together with a fiberwise inner product that can distinguish self-parallel from horizontal variation. Once calM and bbC are available, the minimal structure that supports those operations is an effective complex Hilbert bundle. The bundle is therefore forced by what later derivations require; it is not a new primitive insertion.

2.5.7 $U(1)$ redundancy from

bbC phase

structure 2.5.7 $U(1)$ redundancy from mathbbC phase structure

Theorem 2.5.7 (Local $U(1)$ redundancy as derived from bbC).

Given the complex Hilbert bundle (Theorem 2.5.6) and normalized fibers, the local phase redundancy

$|\psi(x)\rangle \rightarrow e^{i\phi(x)} |\psi(x)\rangle$ with

$C^1(\text{calM}, \text{bbR})$ is the natural redundancy of complex normalization.

Status. Proved in this CF.

Proof. The unit circle in bbC is exactly $U(1) = e^{i\mathbb{R}}$. Acting on a normalized fiber element by $e^{i\phi}$ preserves normalization and changes only phase, not the ray in projective Hilbert space. The local version, where $\phi(x)$ is a C^1 function on calM , is forced by compatibility with the C^1 structure of the representative section. This is not an additional postulate. It is the natural consequence of carrying a complex normalized state on a C^1 carrier.

2.6 Carrier contents at the CF00 stage

The derivation above gives calM enough structure to host local representative variation. At the CF00 stage, calM carries:

- carrier-domain locality from Theorem 2.5.3,
- local differential structure from Theorem 2.5.4,
- carrier-domain distance d_{calM} from the bbR completion metric of Theorem 2.5.1,
- and the minimal carrier topology needed for local representative variation.

The later physical geometry is then induced from the representative ontology built on this carrier. Metric and curvature as physical geometry, the reversible/irreversible dynamical split, support neighborhoods, overlap-based gauge structures, and holonomy observables are developed downstream in the sequence of sections that follow. This keeps calM in its proper role: an earned carrier arena that makes the later quotient-physical structures possible.

2.7 Verification boundary for stage inheritance

The stage inheritance claim is easy to inspect. A reader should be able to follow every structure used in CF00 back to the post-2D stack and the constructions derived from it here. The main places to check are whether any theorem silently requires:

- a primitive support or adjacency graph,
- a primitive constitutive bond or link structure,
- a primitive gauge connection or Wilson datum,
- a primitive metric or curvature tensor,
- or a primitive engine law outside the quotient-geometric construction.

The paper is written so those checks can be made locally. If one of these structures is used, the surrounding theorem should either derive it or explicitly mark it as a later comparison object.

@@HEADING1:3. Carrier Ontology and Representative Structure@@

3.1 Carrier domain

The carrier domain calM is the connected C^1 domain derived in Theorem 2.5.5 from the CF000 invariant stack. Its role at this stage is inherited input in the sense of Theorem 2.4.1: earned, not assumed.

At this stage the carrier is intentionally lean. Support, adjacency, bond structure, and locality-bearing edge sets are not placed into the starting ontology; they are developed later from the representative law.

The only spatial structure used at stage entry is:

enumi.

-
- the derived local differential structure of calM ,
-
- and, when a cone statement is later made, a carrier-domain distance d_{calM} used only to measure that cone.

The distance d_{calM} does not define support. It only measures the region in which a support relation generated by the law must lie.

@@HEADING2:3.2 Effective Hilbert-bundle language@@

By Theorem 2.5.6, the effective complex Hilbert bundle $\pi: \text{calH} \rightarrow \text{calM}$ is the derived fiber structure over calM carrying the representative state-family. The word effective is used literally: each fiber calH_x carries only the minimum inner-product structure needed to define normalization, overlap, projected variation, and local phase redundancy.

This is carrier language. It is not a claim that the effective Hilbert bundle is the final micro-ontology of the universe.

@@HEADING2:3.3 Representative state-family@@

A representative section is a C^1 map
 $\psi: \text{calM} \rightarrow \text{bbS}(\text{calH})$,

$x \mapsto |\psi(x)\rangle$,
 with fiberwise normalization
 $\langle \psi(x), \psi(x) \rangle_x = 1$

for all $x \in \text{calM}$.

The state at this branch level is therefore:

-
 the local normalized representative state-family,
 -
 together with its admissible local variation class.

@@HEADING2:3.4 Local U(1) redundancy3.4 Local U(1) redundancy@@

The representative is redundant under local phase:

$|\psi(x)\rangle \mapsto e^{i\phi(x)} |\psi(x)\rangle$,

$C^1(\text{calM}, \text{bbR})$.

This redundancy is derived in Theorem 2.5.7 from the complex phase structure of the Hilbert bundle. It is not an additional postulate. The physical object is the quotient-physical content that survives removal of the self-parallel phase direction.

@@HEADING2:3.5 Derived representative theorem@@

Theorem 3.5.1 (Derived representative ontology).

The representative state-family on calM together with local $U(1)$ redundancy are the first derived objects of this stage, earned from the CF000 invariant stack via Theorems 2.5.1--2.5.7. Support, adjacency, gauge connection, holonomy, bond structure, observable structure, and spacetime-bearing locality relations are deferred until the sections that derive them from representative variation and support.

Status. Proved in this CF.

Proof. The data listed above are exactly the data needed to define local representative variation and representative redundancy once the carrier, number fields, topology, differentiability, and Hilbert-bundle structure have been earned. Any stronger relational structure such as support graphs, holonomies, constitutive bonds, or gauge link variables would add relational content not yet earned from the representative ontology. Because the purpose of CF00 is to derive those structures, they cannot be inserted primitively without circularity. Therefore the first derived ontology of the stage is exhausted by the representative state-family on calM , the fiberwise inner-product structure, and the derived local $U(1)$ redundancy.

@@HEADING2:3.6 Verification boundary for representative ontology@@

The representative ontology theorem is checked by asking whether any downstream derivation truly requires a primitive insertion of:

- support or adjacency,
- constitutive bond/link structure,
- gauge connection or Wilson data,
- or metric/curvature geometry.

If such a primitive insertion were necessary, the derived representative ontology would be too shallow.

4. Lifted-State Quotient Variation

4.1 Smooth representative projector

Let $\hat{X}$ denote the lifted-state space generated by the Phase Calculus object $\hat{X}_i$ and its admissible operator words.

$\Pi_{\psi} : \hat{X} \rightarrow (\text{calM}, \text{bbP}(\text{calH})),$

which reads a retained lifted state as a ray of normalized complex representatives over calM . The branch law is admissible.

$\Pi_{\psi} \circ E = G_{\psi} \circ \Pi_{\psi}.$

This is the point at which smooth representative language enters. It enters as a projector-visible shadow of retained

4.2 Vertical and horizontal projectors

Choose a normalized local representative $|\psi\rangle$ of the ray $\Pi_{\psi}(\hat{X}_i)$. Define the rank-one projector onto the representative

$P_{\psi} := |\psi\rangle\langle\psi|,$

and the complementary projector by

$P_{\perp} := 1 - |\psi\rangle\langle\psi|.$

For any admissible smooth-shadow variation v of the representative field,

$v = v_{\parallel} + v_{\perp},$

$v_{\parallel} = P_{\psi} v,$

$v_{\perp} = P_{\perp} v.$

The horizontal component $v_{\perp}$ is the smooth branch image of retained lifted variation after pure representative phase

4.3 Why the self-parallel direction is redundant

Under local rephasing,

$|\psi\rangle \rightarrow e^{i\phi} |\psi\rangle,$

the infinitesimal variation generated by changing is proportional to $i|\psi\rangle$. This lies in the representative line spanned by $|\psi\rangle$. It changes the displayed representative but not the lifted branch content read through the physical quotient. In Phase Calculus language, it is visible-witness motion without state-complete content unless retained coordinates change through the lifted operator law.

4.4 Quotient-physical local variation theorem

Theorem 4.4.1 (Horizontal variation as the smooth shadow of retained lifted variation).

The physically meaningful smooth local variation is the horizontal class defined by $P_{\perp}$. The self-parallel direction defined by P_{ψ} is redundant representative motion and must be quotiented out before any physical geometric or dynamical object is constructed.

Proof.

The retained lifted state distinguishes true carried change from visible representative coincidence. The smooth representative projector $P_i\psi$ is lawful only as a quotient descendant of lifted evolution. Local rephasing changes only the representative chosen inside a ray, so its infinitesimal direction lies in the fiber line $bbC|\psi\rangle$. Any bilinear or transport law intended to describe branch-visible physical variation must therefore vanish on this vertical direction. The unique local projection implementing that quotient at the representative level is $P_{\perp}$. Thus the smooth physical variation is the horizontal class $P_{\perp}v$.

4.5 Physical state space at this layer

The physical local state at this layer is not $|\psi\rangle$ itself. It is the retained lifted state read through the ray quotient. The smooth tangent data are therefore horizontal representative variations: local shadows of lifted changes, with vertical phase motion removed.

4.6 Verification boundary for quotient construction

The quotient theorem should be revisited if a vertical representative direction contributes to a gauge-invariant local bilinear object, or if two lifted states with identical smooth ray projection nevertheless force different local geometry after all retained coordinates relevant to the projector are included. Either outcome would mean that the chosen smooth projector is not state-complete for the claimed branch.

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4.7 Retained-state source of horizontal geometry

Phase Calculus closes the finite retained state before any smooth geometry is read out. The carried object is

$\hat{X}_i = (A, q, \theta, \kappa, c),$
 $q = (u, v),$
 $c = [\theta - \frac{\pi}{2}uv, \theta + \frac{\pi}{2}uv],$

and admissible finite evolution is generated by the operator core

$cal U = Q, B, L,$

where Q is quarter continuation, B is balanced refinement, and L is host lift / orthogonal re-articulation. A branch-visible law is accepted only when it descends from retained evolution by an exact quotient criterion

$P_i \circ E = G \circ P_i.$

This criterion is the missing A0 bridge for the geometry: the visible representative is not allowed to decide state identity, and therefore the geometric object cannot be defined from visible phase alone.

Let $P_{i\text{geom}}$ be the smooth carrier-shadow projector that reads the retained two-axis/germ data as a normalized local re-

$\psi(x) \mapsto e^{i(x)}\psi(x),$

because this change alters the representative but not the retained quotient content. Thus every geometry-bearing local

$P_{\perp} = 1 - |\psi\rangle\langle\psi|.$

Theorem 4.7.1 (Retained-state descent forces horizontal variation).

Assume a smooth branch-visible geometry is required to descend from the retained Phase Calculus state by $P_{i\text{geom}} \circ E = G$.

Proof.

The quotient law says branch-visible geometry may depend only on content preserved by the projector. A local rephasing

The next section derives the induced tensor from this retained-state horizontal content. The smooth formula is a notation for the quotient shadow, not an external starting point.

5. Induced Geometry as the Smooth QGT Shadow

5.1 Native quotient derivative

Let x^μ be local coordinates on the derived carrier calM . The raw derivative $_{\mu}|\psi\rangle$ contains both quotient-physical

```
|D_mupsi>
:=
P_perp|mupsi>
=
|mupsi>-|psi><psi|mupsi>.
```

This derivative is the smooth shadow of lifted-state variation after the phase-memory direction has been quotiented out.

5.2 From retained response to the familiar formula

The retained-state quotient supplies two local invariants before the QGT is named. Nearby representatives have a quotient

```
cal D(x,x+dx)
:=
1-|<psi(x),psi(x+dx)>|^2,
```

whose leading nonzero coefficient is symmetric. A small oriented coordinate loop carries a quotient-visible phase holonomy

```
Q^VDM_munu
:=
< D_mupsi D_nupsi>.
```

Expanding $D_\mu = P_\perp \mu$ gives

```
Q^VDM_munu
=
< _mupsi _nupsi>
-
< _mupsi>
<psi _nupsi>.
```

This is the familiar quantum-geometric tensor expression. In CF00 the formula is written only after the retained-state

5.3 Gauge invariance of the lifted response

The proof of Theorem 4.7.1 already shows that the projected derivative is invariant up to the common representative phase

```
Q^VDM_munu
< e^iD_mupsi e^iD_nupsi>
=
Q^VDM_munu.
```

The response tensor is therefore gauge-invariant because it is horizontal, not because gauge geometry was inserted before

5.4 Induced-QGT theorem

Theorem 5.4.1 (QGT as smooth quotient descendant).

Given the CF000 invariant stack, the Phase Calculus lifted state and operator core, the derived carrier calM , the effective

```
Q_munu:=Q^VDM_munu
=
< D_mupsi D_nupsi>
```

is the smooth quotient descendant forced by the retained-state grammar. It is not an assumed classical starting point.

Proof.

The retained grammar forces the quotient to preserve state-complete transport, refinement magnitude, and oriented orthogonality

5.5 Metric and curvature split

Define

```
g_munu:=ReQ_munu,
```

```
Omega_munu:=-2ImQ_munu.
```

Then

$$Q_{\text{munu}} = g_{\text{munu}} - \frac{1}{2} \Omega_{\text{munu}}.$$

The tensor g is real and symmetric. The tensor Ω is real and antisymmetric.

Theorem 5.5.1 (Metric and curvature from lifted burdens).

The real sector g is the smooth response to finite-resolution refinement, and the antisymmetric sector Ω is the smooth response to oriented orthogonal transfer. Neither sector is primitive.

Proof.

A Hermitian tensor decomposes uniquely into real symmetric and imaginary antisymmetric sectors. By Theorem 5.4.1, the Hermitian tensor is a retained-state quotient descendant. Its real sector carries the phase-insensitive overlap-loss / refinement-magnitude response, while its imaginary sector carries the oriented loop / quarter-transfer response. Therefore both sectors are descendants of the retained-state quotient geometry.

5.6 A hard-checkable symbolic example

The lifted-state-to-QGT construction has a compact local chart that can be checked directly. In the representative sha

$$|\psi(u, v)\rangle = \frac{1}{\sqrt{1+u^2+v^2}}$$

$$\begin{pmatrix} 1 \\ u+iv \end{pmatrix}, \quad u, v \in \mathbb{R},$$

the horizontal covariance formula gives

$$Q(u, v) = \frac{1}{(1+u^2+v^2)^2}$$

$$\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}.$$

Therefore

$$g = \frac{1}{(1+u^2+v^2)^2}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$\Omega = \frac{1}{(1+u^2+v^2)^2}$$

$$\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}.$$

The determinants reduce to

$$g = \frac{1}{(1+u^2+v^2)^4},$$

$$\Omega = \frac{4}{(1+u^2+v^2)^4},$$

$$\frac{g}{\Omega} = \frac{1}{4}.$$

This is a small but useful numerical anchor for the paper: the local chart exhibits the quotient-induced metric/curvat

5.7 Verification boundary for induced geometry

The induced-geometry chain should be revisited if the smooth representative projector fails the quotient law, if the projected derivative is not gauge-invariant, if the horizontal covariance depends on vertical representative content, if the split into g and Ω is not uniquely induced from Q , or if later dynamics require geometric tensors not earned from horizontal lifted-state

covariance.

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6. Horizontal Transport and Emergent Dynamical Architecture

@@HEADING2:6.1 Tangent decomposition of representative evolution@@

Let $t \psi(t)$ be an admissible representative trajectory.

Since

$$\langle \psi, \psi \rangle = 1,$$

differentiate in time:

$$\frac{d}{dt} \langle \psi, \psi \rangle$$

$$=$$

$$\langle \dot{\psi}, \psi \rangle + \langle \psi, \dot{\psi} \rangle$$

$$=$$

$$2 \operatorname{Re} \langle \dot{\psi}, \psi \rangle$$

$$=$$

$$0.$$

Hence $\langle \dot{\psi}, \psi \rangle$ is purely imaginary. Therefore there exists a real scalar functional such that

$$\langle \dot{\psi}, \psi \rangle = -i \lambda.$$

Define

$$\tilde{\psi} := \dot{\psi} + i \lambda \psi.$$

Then

$$\langle \tilde{\psi}, \tilde{\psi} \rangle$$

$$=$$

$$\langle \dot{\psi}, \dot{\psi} \rangle + i \langle \dot{\psi}, \psi \rangle + i \langle \psi, \dot{\psi} \rangle$$

$$=$$

$$-i \lambda + i \lambda$$

$$=$$

$$0.$$

Therefore every norm-preserving representative evolution decomposes uniquely as

$$=$$

$$-i \lambda \psi + \tilde{\psi},$$

$$\langle \tilde{\psi}, \tilde{\psi} \rangle = 0.$$

@@HEADING2:6.2 Interpretation of the decomposition@@

The term $-i \lambda \psi$ is vertical. It is pure representative phase motion. The term $\tilde{\psi}$ is horizontal. It is the quotient-physical evolution class.

@@HEADING2:6.3 The geometric operators available for dynamics@@

Once the quotient geometry has been induced, there are exactly two geometric structures available at this level:

-
the antisymmetric sector Ω ,
-
the symmetric sector g .

On the regular quotient sectors these define bundle maps from cotangent data to tangent data:
 $J := \Omega^\wedge,$
 $M := g^\wedge.$

This is the first appearance of plain M in the document. It is the dissipative symmetric operator induced from the quotient geometry. It is not the carrier domain.

6.4 Why the mixed law is forced

The finite lifted grammar separates retained transport from visible projection. After projection to the smooth carrier
 $J := \Omega^\wedge,$
 $M := g^\wedge.$

The antisymmetric sector preserves ORS order and supplies reversible horizontal transport. The symmetric sector measures
This is the dynamical reading of the retained-first derivation. The lifted operators decide lawful state transport. The smooth shadow reads that transport through horizontal covariance. The two covariance sectors then provide the first reversible and irreversible channels available to the projected branch.

6.5 Emergent engine theorem

Theorem 6.5.1 (Emergent scalar-generated mixed dynamical architecture).
Let the quotient geometry be given by the induced pair (g, Ω) .
Suppose the horizontal law is required to satisfy all of the following:
enumi.

- it is local on $\text{cal}M$,
- it acts only on quotient-physical data,
- it is generated by scalar state functionals,
- it contains both the reversible sector carried by the antisymmetric induced geometry and the irreversible sector carried by the symmetric induced geometry,
- it does not insert any further primitive generator tensor not already earned from the induced geometry.

Then the horizontal law must take the scalar-generated mixed form
 $= J \, d\text{cal}I + M \, d\text{Sigma}$
 $=$
 $\Omega^\wedge \, d\text{cal}I + g^\wedge \, d\text{Sigma},$
for scalar generators $\text{cal}I$ and Sigma .

Proof.
Condition (2) excludes vertical representative content. Condition (5) excludes any new primitive tensor beyond the induced geometric data. Thus the only available local bundle maps from scalar differentials to quotient tangent vectors are those induced from Ω and g .

An antisymmetric bilinear structure acting on a scalar differential gives the reversible local transport allowed by the induced curvature sector. A symmetric bilinear structure acting on a scalar differential gives the gradient-like local transport allowed by the induced metric sector. If both sectors must appear and both must be scalar-generated, then every admissible horizontal law is the sum of one term generated by the antisymmetric operator and one term generated by the symmetric operator. No third independent primitive sector may appear by condition (5). Therefore the general scalar-generated law has the form

$$= J \, d\alpha I + M \, d\sigma.$$

Since $J=\Omega$ and $M=g$ on the regular quotient sectors, this is exactly

$$= \Omega \, d\alpha I + g \, d\sigma.$$

Thus the dynamical engine is not imported. It is forced by the induced quotient geometry under the stated conditions. \$\$

6.6 Strength of the result statement

The theorem above is a stage result. It derives the first smooth transport architecture visible after retained lifted evolution is read through the horizontal covariance branch. It does not claim that all finite retained-state dynamics have been reduced to a differential equation. It says that whenever the smooth carrier branch is the active projector, admissible scalar-generated dynamics must use the induced $J+M$ channels and must remain vertical-blind.

6.7 Concrete dynamical consequence: vertical-blind transport

If two representative velocities differ only by local phase velocity,

$$v' = v + i, \\ R,$$

then

$$P_{\text{perp}}' = P_{\text{perp}} + i \, P_{\text{perpsi}} = P_{\text{perp}}.$$

Therefore the horizontal covariance Q , its sectors g and Ω , and the scalar-generated law $Jd\alpha I + Md\sigma$ are insensitive to vertical phase shifts.

This is the first concrete dynamics result of CF00. Geometry is not merely a label on a state space; it closes a transport channel. Reversible motion must pass through the Ω -sector, irreversible motion must pass through the g -sector, and both act only on horizontal quotient data. When the full CF01 metriplectic degeneracy conditions are imposed downstream, this same split becomes the energy/entropy-compatible $J+M$ engine. CF00 supplies the lifted-state-derived smooth geometry and the vertical-blind transport restriction that the later engine uses.

The standalone SymPy residual script attacks the local algebraic form of this statement by checking that projected derivatives and recovered-QGT contractions are unchanged under symbolic vertical phase shifts in the Section 5.6 chart.

@@HEADING2:6.8 Verification boundary for emergent engine@@

The emergent-engine theorem is checked by asking whether any of the following are required:

- a primitive J -sector or M -sector is needed,
- a third primitive dynamical sector is needed,
- or the induced geometry cannot generate both reversible and irreversible channels.

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@@HEADING1:7. Constitutive Narrowing and Reservoir Equivalence@@

@@HEADING2:7.1 Need for a constitutive sector@@

The mixed law from Section 6 is structural. To obtain a concrete physical engine one must specify the invariant generator call and entropy generator Sigma.

@@HEADING2:7.2 Minimal conservative/entropy-bearing form@@

The narrowest admissible constitutive form at this level is

$$\begin{aligned} \text{calI}[\psi, u] &= \\ &= \int_{\text{calM}} \left(\frac{1}{2} |D_{\text{t}\psi}_{\text{perp}}|_g^2 + \frac{1}{2} |D_{\text{i}\psi}_{\text{perp}}|_g^2 + V([\psi]) + u \right) d\mu, \\ &\text{and} \\ \text{Sigma}[\psi, u] &= \int_{\text{calM}} s(u) d\mu, \\ &\text{with} \\ s &\in C^2, \\ s'(u) &> 0, \\ s''(u) &< 0. \end{aligned}$$

Here:

- $V([\psi])$ is a local gauge-invariant quotient potential,
- and D determine the telegraph/Cattaneo principal part,
- u is the local entropy-bearing reservoir variable,
- and $d\mu$ is the carrier-domain integration measure.

@@HEADING2:7.3 Why the reservoir is required@@

The irreversible sector cannot be closed on the conservative variables alone without collapsing the distinction between the invariant generator and the entropy generator. The reservoir variable is therefore not ornamental. It is the minimal additional scalar field needed to carry the irreversible channel while preserving that distinction.

@@HEADING2:7.4 Reservoir-coordinate conjugacy lemma@@

Let

$$\begin{aligned} \text{calI}[\psi, u] &= \\ &= \text{calI}_0[\psi] + \int_{\text{calM}} u d\mu, \\ \text{Sigma}[\psi, u] &= \int_{\text{calM}} s(u) d\mu, \end{aligned}$$

where $\text{calI}_0[\psi]$ is the conservative sector and $s'(u) > 0$ on the reservoir range. Define

$$\begin{aligned} &: (\psi, u) \rightarrow (\psi, s(u)), \\ &= s(u). \end{aligned}$$

Lemma 7.4.1 (Reservoir-coordinate conjugacy).

On every region where $s'(u) > 0$, the map is a local C^2 diffeomorphism on the reservoir sector. In the transformed coordinates one has

$$\text{cal } I[\psi,] = \text{cal } I_0[\psi] + \text{cal } M s^{-1}(\cdot) \text{ d}\mu,$$

$\text{Sigma}[\psi,]$

$= \text{cal } M \text{ d}\mu.$

The primitive law in (ψ, u) and the primitive law in $(\psi, \cdot)$ are locally conjugate. Their degeneracy content, entropy-production sign, derived support relation, and runtime-admissibility content are the same physical content written in different reservoir charts.

Proof.

Because $s'(u) > 0$, the inverse function theorem gives a local inverse $u = s^{-1}(\cdot)$, so is a local diffeomorphism.

Let X be the primitive law vector field in (ψ, u) coordinates.

The transformed law is

$X = T X^{-1}.$

Thus the two laws are dynamically conjugate.

The degeneracy statements are kernel statements for the induced symmetric and antisymmetric operators acting on the differentials of the scalar generators. Since those tensors and differentials transform covariantly under $\cdot$, the vanishing or nonvanishing of those kernel conditions is unchanged. Therefore the degeneracy content is preserved.

The transformed entropy is

$\text{Sigma}[\psi,]$

$=$

$\text{Sigma}[\psi, s^{-1}(\cdot)].$

Hence along corresponding trajectories one has

fracddtSigma

$=$

$\text{fracddtSigma}.$

So the sign of entropy production is unchanged.

If U_{t,t_0} is the nonlinear solution operator, then

$U_{t,t_0} = \circ U_{t,t_0} \circ \wedge^{-1}.$

Differentiating gives

$D U_{t,t_0}$

$=$

$D \circ D U_{t,t_0} \circ D \wedge^{-1}.$

Therefore the support relation defined by nonzero propagated horizontal perturbations is preserved. Since runtime admissibility is defined only by faithful discretization of the carrier, quotient, law, and derived support relation, the runtime-admissibility content is also preserved. $\$ \$$

@@HEADING2:7.5 Constitutive equivalence

theorem@@

Theorem 7.5.1 (Theorem-grade constitutive equivalence class).

The entropy-bearing constitutive sector is theorem-grade at the level of the equivalence class

$s \in C^2,$

$s'(u) > 0,$

$s''(u) < 0,$

modulo smooth strictly monotone reservoir-coordinate reparameterization.

Proof.

By Lemma 7.4.1, any two entropy charts related by such a

reparameterization are locally conjugate, preserve the mixed law structure, preserve the degeneracy content, preserve entropy-production sign, preserve the support theorem content, and preserve runtime-admissibility content. Therefore the theorem-grade constitutive content is the equivalence class, not any one special chart. \$\$

@@HEADING2:7.6 Canonical execution specialization@@

A canonical execution specialization is
 $s(u) = k_B u$,
 $u > 0$.

This specialization belongs to the theorem-grade equivalence class but is not uniquely forced by the theorem.

@@HEADING2:7.7 Verification boundary for constitutive narrowing@@

The constitutive equivalence theorem is checked by applying smooth strictly monotone reservoir-coordinate changes and tracking whether they alter:

- the mixed-law structure,
- degeneracy content,
- the sign of entropy production,
- the derived support relation,
- or the runtime-admissibility content.

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@@HEADING1:8. Derived Locality, Support, and Gauge-Hosting@@

@@HEADING2:8.1 Nonlinear solution operator@@

Let Z denote the primitive state space of admissible fields (ψ, u) modulo local $U(1)$ redundancy on the representative sector. Let

$U_{t,t_0}: Z \rightarrow Z$

denote the nonlinear solution operator of the closed primitive law on its interval of existence.

For each admissible trajectory $z()$ and each pair t, t_0 ,

let

$D U_{t,t_0}|_z(t_0)$

denote the Frchet derivative of the solution operator at the initial state.

@@HEADING2:8.2 Derived support relation@@

A horizontal perturbation δz_0 at time t_0 is localized near x_0 calM if its support lies in an arbitrarily small

neighborhood of x_0 in the primitive carrier domain.

```

Define
(x_0,t_0) \squigarrow (x,t)
  if and only if there exists a localized horizontal perturbation
  delta z_0 near x_0 such that
  (D U_t,t_0|_z(t_0) delta z_0)(x) != 0.

Define the exact support set by
Supp(t;t_0,x_0)
=

x calM :
(x_0,t_0) \squigarrow (x,t)
.

```

8.3 Derived-support theorem

Theorem 8.3.1 (Derived support from the law).

For every admissible trajectory of the closed primitive law, the locality-bearing support relation is the support relation induced by the retarded propagation of horizontal perturbations under the Frchet derivative of the nonlinear solution operator. It is not primitive adjacency, not thresholded observable recovery, and not graph structure inserted by hand.

If the principal part of the closed primitive law is of telegraph/Cattaneo type with effective characteristic speed

c_{eff} , then

$\text{Supp}(t;t_0,x_0)$

subset eq

$x \text{ calM} :$

$d_{\text{calM}}(x,x_0) \leq c_{\text{eff}}(t-t_0)$

for all $t \geq t_0$ within the domain of existence.

Proof.

The support relation is defined entirely through the law via its nonlinear solution operator and the propagation of horizontal perturbations under its Frchet derivative. Therefore support is generated by the law itself.

Because the perturbations are horizontal, representative phase redundancy does not affect whether a perturbation propagates nontrivially. So the support object is quotient-physical.

If the principal part of the law is telegraph/Cattaneo type, then the domain of dependence of perturbations is bounded by the corresponding characteristic cone measured with the carrier-domain distance d_{calM} . Therefore the derived support set lies inside that cone. $\square$

@@HEADING2:8.4 Observable-independence

corollary@@

Corollary 8.4.1 (Observable-independence).

Let $\text{cal } F$ be any separating family of local gauge-invariant observables on the horizontal quotient dynamics. Then $\text{cal } F$ witnesses the derived support relation but does not define it. The

support set $\text{Supp}(t; t_0, x_0)$ is independent of the separating family used to detect propagated perturbations.

Proof.

The support object is defined from the law itself through $D U_{t, t_0}$. Any separating witness family merely detects whether the propagated perturbation is nonzero. It does not alter the propagation. Therefore different separating witness families recover the same support object. $\$ \$$

8.5 Non-circularity lemma

Lemma 8.5.1 (Non-circularity of derived support).

The primitive inputs of the formalism are the carrier domain calM , the representative field, the reservoir field, and the closed primitive law on those fields. No support relation, adjacency graph, or neighborhood graph on calM is primitive. The support relation on calM is derived from the retarded propagation of horizontal perturbations under the law.

Proof.

The only primitive role of calM is to provide the arena on which fields are defined. The support relation is a subset of $\text{calM} \times \text{calM} \times \mathbb{R}^2$ selected by nonzero propagation under the solution operator. Since that relation is computed from the law and is not independently specified, no primitive locality-bearing adjacency is inserted. $\$ \$$

@@HEADING2:8.6 Local overlap cover and loop-hosting lemma@@

Let $\psi: \text{calM} \rightarrow \mathbb{R}$ be continuous and normalized. For fixed $x \in \text{calM}$, the map $y \mapsto \langle \psi(x), \psi(y) \rangle$ is continuous and equals 1 at $y=x$.

Lemma 8.6.1 (Local overlap cover and loop-hosting).
 For every $x \in \text{calM}$ there exists an open neighborhood $U_x \subset \text{calM}$ such that $\langle \psi(x), \psi(y) \rangle \neq 0$
 for all $y \in U_x$.
 Hence the family $U_x \subset \text{calM}$ is an open overlap cover on which local overlap phases $\text{cal } U(x, y)$
 $=$
 $\frac{\langle \psi(x), \psi(y) \rangle}{|\langle \psi(x), \psi(y) \rangle|}$
 are well-defined whenever x and y lie in a common overlap neighborhood.

If a support-generated neighborhood $V \subset \text{Supp}(t; t_0, x_0)$ is contained in one such overlap neighborhood, then:

enumi.

-
- local link phases on V are well-defined and gauge-covariant;
-
- any sufficiently small loop in V admits a discrete holonomy built

from ordered products of local link phases;

-

plaquette and Wilson-loop constructions on such sufficiently small support-generated loops are well-defined.

Proof.

Continuity of the overlap map and normalization at coincidence imply the existence of a neighborhood U_x on which the overlap remains nonzero. The normalized overlap phase is therefore well-defined on each such neighborhood. Under local $U(1)$ rephasing, the normalized overlap phase transforms covariantly and therefore defines a local link variable.

If a support-generated neighborhood is subordinate to the overlap cover, then any sufficiently small loop in that neighborhood is covered by finitely many overlap neighborhoods. Ordered products of local link variables along the loop are therefore defined, and these products give local plaquette and Wilson-loop holonomies. \$\$

@@HEADING2:8.7 Gauge-hosting sufficiency
corollary@@

Corollary 8.7.1 (Gauge-hosting sufficiency of derived support).

Within the scope of CF00, the derived support neighborhoods furnished by the law are sufficient to host the downstream local overlap-based connection, plaquette, and Wilson-loop machinery. Therefore gauge-holonomy structure in this branch depends on derived support and continuous representative geometry, not on inserted primitive adjacency.

Proof.

By Lemma 8.6.1, each point admits an overlap neighborhood with well-defined local link phases. By Theorem 8.3.1, the law furnishes local support neighborhoods. On sufficiently fine local scales the support neighborhoods may be taken subordinate to the overlap cover. Therefore the local overlap-based gauge machinery is structurally supported on derived support. \$\$

@@HEADING2:8.8 Verification boundary for derived locality and gauge hosting@@

The support theorem or its corollaries should be revisited if:

-

support cannot be defined directly from the solution operator and its Fr-chet derivative,

-

different separating witness families define inequivalent support objects,

-

support requires primitive adjacency,

-

support violates the carrier-domain cone bound,

-

or support-generated neighborhoods do not admit local overlap-based gauge constructions.

9. Runtime Admissibility

A runtime architecture is admissible relative to CF00 on a finite validation horizon if it satisfies the following sufficient conditions.

Theorem 9.1 (Runtime admissibility).

Let $\text{cal } R_h$ be a discretized implementation. Suppose the following hold on a finite validation horizon.

enumi.

-
- $\text{cal } R_h$ discretizes the primitive carrier variables (ψ, u) on $\text{cal } M$ and preserves normalization and the local $U(1)$ quotient structure up to bounded residuals.
-
- $\text{cal } R_h$ discretizes the closed scalar-generated primitive law together with the theorem-grade constitutive equivalence class of the entropy-bearing reservoir sector.
-
- $\text{cal } R_h$ computes support from the discrete retarded propagator of the discretized law and does not insert adjacency, support, or neighborhood graphs as primitive ontological inputs.
-
- $\text{cal } R_h$ preserves, up to bounded residuals on the validation horizon,
-
- induced-QGT regularity,
-
- the metric/curvature split,
-
- the mixed generator structure,
-
- entropy-production sign,
-
- cone inclusion of derived support relative to $d_{\text{cal } M}$,
-
- and the overlap-cover conditions needed for local overlap-based gauge constructions.
-
- $\text{cal } R_h$ introduces no undeclared conserved quantities, primitive support structures, or gauge inconsistencies beyond those allowed by the formalism.

Then $\text{cal } R_h$ is an admissible runtime realization of CF00 on that validation horizon.

Proof.

Sections 2 through 7 determine the ontologically load-bearing structures of this branch: primitive representative carrier, quotient structure, induced geometry, scalar-generated mixed law, constitutive equivalence class, derived support relation, and local gauge-hosting sufficiency. A

runtime satisfying conditions (1) through (5) preserves exactly those structures, up to bounded residuals on the stated horizon, and keeps the additional structures within the ontology derived in this paper. Therefore it is an admissible realization of CF00 on that horizon. \$\$

This theorem gives a clear sufficiency statement for implementations: make the listed structures and residuals visible, preserve them on the declared horizon, and the realization belongs to the CF00 retained-state carrier-geometry class.

@@HEADING2:9.1 Verification boundary for runtime
admissibility@@

A concrete implementation is outside this admissibility theorem when it:

- inserts primitive adjacency,
- does not preserve the quotient or induced-geometry structure,
- violates the carrier-domain cone bound,
- or introduces hidden integrals or primitive support structures not licensed by the formalism.

The executable notebook illustrates these checks on representative examples and records the measured residuals. The proof burden remains in the paper.

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10. Validation and Reproducibility Notes

The main text carries the definitions, theorem statements, the A0-closure source map in Section 2.4A, and the hard-checkable symbolic example in Section 5.6. The companion notebook gives readers a practical way to inspect the construction numerically: it checks representative completions, quotient projections, gauge-invariant quantities, mixed-law decompositions, vertical-blind transport, constitutive conjugacy, and support-locality behavior. The companion package also includes a standalone SymPy script for the algebraic residuals and a Lean 4 theorem-surface file for the exact algebraic and dependency-order claims. These examples are intentionally small so they can be rerun, modified, and reused.

10.1 Primitive carrier diagnostics

The carrier derivation is most directly checked by tracing each carrier ingredient back to the inherited stack and the completion steps in Section 2. A useful diagnostic is to ask whether adjacency, gauge data, constitutive bonds, or observables appear before they have been derived.

10.2 Induced geometry diagnostics

The induced-geometry chain is checked by following the lifted-state descent: visible projection should fail to be state-complete, vertical representative content should drop out, the native horizontal tensor should recover the QGT expression in a smooth chart, and the metric/curvature split should come from that Hermitian response rather than from an added geometric object.

10.3 Mixed-law diagnostics

The mixed-law section is checked by verifying that the reversible and irreversible channels are generated from the induced geometry and scalar reservoir structure. The natural comparison is a construction that keeps only one channel; such a comparison shows why both sectors are needed for the later metriplectic bridge.

10.4 Constitutive equivalence diagnostics

The constitutive equivalence theorem is checked by changing reservoir coordinates and confirming that the mixed law, degeneracy content, entropy-production sign, derived support relation, and runtime-admissibility content are preserved under the conjugacy.

10.5 Derived support diagnostics

The support theorem is checked through the nonlinear solution operator and its Frechet derivative. Observables should witness one support relation rather than define incompatible support objects, and the resulting support should remain inside the carrier-domain cone bound.

10.6 Runtime diagnostics

A runtime realization belongs to the CF00 admissibility statement when it preserves the quotient and induced-geometry structures, respects the cone bound, and avoids hidden primitive adjacency or support structures. The notebook records these checks as reusable numerical examples rather than as replacements for the formal derivation.

10.7 Companion verification materials

The companion package contains three reusable verification surfaces. The notebook records the executable numerical examples.
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11. Canon Integration

CF00 now sits inside the retained-first canon rather than merely pointing to it. CF000 supplies the primitive invariant stack. Phase Calculus supplies the exact lifted-state grammar and quotient criterion. CF00 supplies the smooth induced-geometry branch of that grammar. CF01 then supplies the full QGT-to-metriplectic bracket formalism.

11.1 CF01

CF01 is the downstream bracket-completion paper. CF00 derives the horizontal covariance and its sectors (g, Ω) ; CF01 continues from those sectors to the full metriplectic degeneracy and bracket architecture.

11.2 CF11

CF11 is a downstream derived-limit module. Its scalar-generated law form and entropy-bearing reservoir logic remain consistent with CF00, but CF00 now supplies the smooth geometry and vertical-blind transport restriction underneath that use.

11.3 Other CFs

CF03, the rewritten Infinity Resolution branch, Farey remainder recursion, Transcendental Wall, Operational Utility, and the later Phase Calculus modules are not ornamental references. They are downstream or parallel witnesses of the same retained-first method: unresolved burden is retained in lifted state, saturation forces lawful re-hosting, and visible branch laws are accepted only by quotient descent.

11.4 Phase Calculus and lifted-state integration

Phase Calculus is the finite retained-state substrate used by the revised CF00 derivation. Its primitive carried object is

$\hat{X}_i = (A, q, \theta, \kappa, c),$

its finite operator core is $Q, B, L,$ and its exact branch law is the quotient criterion

$\Pi \circ E = G \circ \Pi.$

CF00 is the smooth horizontal-covariance branch of that criterion. The induced pair (g, Ω) is what becomes visible when the lifted state is projected to normalized complex representatives, vertical phase motion is quotiented out, and the remaining

horizontal variation is measured by its minimal Hermitian covariance.

This placement gives the paper its load-bearing role. The geometry used by CF01 and later modules is not a classical object adopted at the carrier stage. It is a smooth quotient shadow of retained lifted transport. Later branch-memory, hierarchy, monodromy, inverse-fiber, multiscale, and operational-utility papers can therefore reuse (g,Ω) as the smooth branch geometry while keeping the full lifted-state grammar underneath

it.[Lietz2026PhaseCalculus,Lietz2026QuotientDescent,Lietz2026OperationalUtility]

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12. Internal Canon References

The following internal canon modules are retained as provenance, downstream placement, or consistency witnesses. They support orientation and reuse while the derivation itself is carried in this paper.

- Phase Calculus --- direct retained-state operator grammar, exact quotient criterion, and smooth-shadow substrate for CF00.
- CF01 --- provenance and downstream effective engine placement.
- CF02 --- provenance and downstream contact/GENERIC compatibility witness.
- CF04 --- provenance and downstream finite-speed / cone witness.
- CF05 --- provenance and downstream closure / no-hidden-integrals witness.
- CF06 --- provenance and downstream Fisher / information-geometric witness.
- CF09 --- provenance and downstream overlap/holonomy witness.
- CF11 --- provenance and downstream derived-limit reservoir/entropy witness.
- Quotient Descent --- exact branch-visibility criterion and projector-preserved law surface.
- Operational Utility --- multiscale invariance and practical retained-state utility surface.

Where this document reproduces, absorbs, or strengthens structures historically associated with those documents, the proof burden is carried here and not delegated back to them.

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@@HEADING1:13. External Mathematical
Background@@

The following external mathematical categories provide familiar language and comparison points for readers.

enumi.

- Projective Hilbert space and geometric quantum mechanics

Used for representative-to-ray quotient language, vertical/horizontal decomposition, and local overlap geometry.

-

Quantum geometric tensor, Fubini--Study, and Berry geometry

Used as familiar comparison language for the horizontal covariance tensor and its induced split into metric and curvature sectors after the lifted-state derivation has been carried.

-

GENERIC, contact, and metriplectic structure

Used for reversible/irreversible mixed-generator language and entropy-production form.

-

Hyperbolic / telegraph propagation and differentiable

nonlinear flow operators

Used for the support theorem through the nonlinear solution operator and its Fréchet derivative.

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14. Novel Contributions of CF00

The following are the main retained-state carrier-geometry contributions formalized here.

- The retained-first A0-closure source map: CF000 primitives, Phase Calculus lifted grammar, CF00 smooth shadow, and d
- The derivation of the QGT as the smooth horizontal covariance shadow of retained lifted-state transport rather than
- The integration of carrier derivation, quotient-physical variation, horizontal covariance, and induced (g, Ω) sec
- The hard-checkable local chart where the induced sectors satisfy $(g)/(\Omega)=1/4$ exactly.
- The derivation of vertical-blind transport as the first concrete dynamical consequence of the geometry.
- The scalar-generated J+ M architecture as the first smooth dynamical law available from the induced sectors.
- The theorem-grade constitutive equivalence class under reservoir-coordinate conjugacy.
- The support theorem formulated directly from the nonlinear solution operator and its Frechet derivative.
- The explicit non-circularity theorem for derived support.
- The local overlap-cover and loop-hosting lemma proving gauge-hosting sufficiency of derived support neighborhoods.
- The runtime admissibility theorem stated as a sufficient retained-first realization condition.
- The direct canon integration with Phase Calculus, Quotient Descent, CF01, and later retained-state modules.
- The derivation of bbR , bbC , topology, the primitive half-turn constant π , C^1 differentiability, the carrier domain

0.50.5pt

15. Scope and Open Directions

CF00 is stage-complete for the smooth induced-geometry branch developed here. It derives the carrier, the smooth representative quotient, the horizontal covariance tensor, the induced sectors (g, Ω) , and the first vertical-blind transport law from CF000 plus the Phase Calculus retained-state grammar.

The title's "emergent dynamics" should be read in this exact stage-local sense. CF00 derives the admissible transport architecture visible in the smooth branch: vertical-blind horizontal motion and the scalar-generated J+ M split. It does not claim to finish every finite retained-state runtime law, nor does it replace CF01's full bracket-degeneracy construction. It supplies the geometry and first transport restriction that those later modules use.

Natural continuations are now sharper: full Lean formalization of the horizontal covariance uniqueness theorem, deeper finite retained-state derivations of concrete smooth projectors, explicit QGT-to-metriplectic brackets, runtime validations, and additional quotient-shadow branches for hierarchy, branch memory, inverse-fiber states, and multiscale utility.

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@@HEADING1:16. Stage Status of the Carrier Domain

M@@

The question answered here is not whether deeper metaphysics can be imagined. The question is whether the formalism, at its present stage,

forces anything deeper than the carrier domain already derived in Section 2.5.

16.1 What M is in CF00

calM is the derived carrier arena of the branch. It is the connected C^1 domain on which the representative state-family and reservoir field live.

16.2 What M prepares but does not pre-load

calM is not:

- - a primitive metric geometry,
 -
 - a primitive gauge connection,
 -
 - a primitive support graph,
 -
 - a primitive bond structure,
 -
- or the mixed dynamical engine.

@@HEADING2:16.3 What is already encoded in M@@

What is already encoded in calM at this stage is only what has been derived before use: coordinate-bearing real completion, complex fiber compatibility, topology, C^1 variation structure, and the minimal locality needed to state local variation and cone bounds.

16.4 Stage-status statement

Within CF00, calM is the inherited derived input --- derived in Section 2.5 from the CF000 invariant stack, and then serving as the carrier arena from which every later load-bearing structure in this document is induced. The use of bbC, topology, differentiability, and the carrier itself is staged explicitly before those structures are used. All are earned before being used.

@@HEADING2:16.5 Verification boundary for stage status@@

The claim that calM is the correct carrier stage for CF00 is checked by asking whether any theorem requires:

- - a deeper substrate to define representative redundancy,
 -
 - a deeper substrate to define projected variation,
 -
 - a deeper substrate beyond the lifted-state grammar to derive the QGT shadow,
 -
 - a deeper substrate to derive the mixed law,
 -
- or a deeper substrate to derive support.

No such requirement is used in this branch beyond the retained lifted-state grammar explicitly named in Sections 2.4A--2.4B, so the stage-status claim is preserved.

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17. Status and Reuse Note

The theorem-grade content carried here is the derivation of the carrier from the post-2D invariant stack, the retained-first A0-closure source map, the lifted-state descent spine, the representative ontology, quotient-physical horizontal variation, QGT as smooth horizontal covariance shadow, metric/curvature split, tangent decomposition, vertical-blind transport, scalar-generated mixed law, constitutive equivalence under reservoir-coordinate conjugacy, derived support from the nonlinear solution operator and its Frechet derivative, observable-independence, non-circularity, local overlap-cover / loop-hosting, gauge-hosting sufficiency, and runtime admissibility.

17.1 Lean and SymPy status

The companion package includes `lean/CF00_core.lean`, a Lean 4 theorem-surface file for the algebraic, dependency-order, and horizontal-quotient core. It formalizes theorem-shaped versions of the completion-chain dependency, ORS quarter-turn algebra, the continuous-ORS interface, the retained-first source map, the Phase Calculus-to-CF00 quotient path, horizontal projection, and vertical-blind transport. The file deliberately treats Cauchy completion, topological continuity, Hilbert-bundle analysis, and the full uniqueness proof of Hermitian covariance at the interface level rather than claiming a complete mathlib-backed analytic formalization. It is therefore a Lean sketch surface for the exact claims currently feasible at small scale, with deeper analysis-library formalization left as a natural continuation.

The package also includes `sympy/CF00_residual_checks.py`, which independently verifies the Section 5.6 horizontal-covariance identities, determinant identity, vertical-annihilation residuals, and vertical-shift invariance. The script records exact symbolic residual zero in `sympy/CF00_residual_summary.json`.

The natural downstream work is executable realization, numerical witness generation, figure production, bounded-residual checks for concrete implementations, full Phase Calculus projector integration, and the explicit QGT-to-metriplectic bracket mapping developed in CF01. CF00 is therefore complete as the current retained-state-to-smooth-geometry formalism for this branch, and it is designed to be reused as the stage reference for those later constructions.

18. References and Provenance

The theorem-bearing derivation of CF00 is carried in Sections 2--10. The references listed here are divided into direct inheritance, downstream canon alignment, and external mathematical background. Downstream and external items are included for integration and reader orientation; the derivations used in CF00 are stated in the body of the paper.

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